

Roll No. [Total No. of Pages : 4

CY-1844

B. Tech. (First Semester)

EXAMINATION, 2025-26

ENGG. CHEMISTRY

Time : Two Hours

Maximum Marks : 100

Note : Attempt questions from all Sections as directed.

Section—A

(Very Short Answer Type Questions)

Note : Attempt any *seven* questions. : Each question carries 8 marks. $7 \times 8 = 56$

1. Write the full form of VSEPR theory and give its *two* limitations.
2. Write the law of constancy of interfacial angles.

P. T. O.

3. Define rate of reaction. What are its units ?
4. Explain the term 'Degree of Freedom'.
5. Give two examples of organic substitution reactions.
6. Why is dichloroacetic acid stronger than monochloroacetic acid ?
7. What is meant by vulcanisation ?
8. What are thermoplastic and thermosetting resins ?
9. Explain the term 'chromophores'.
10. What is Biogas ?

Section—B

(Short Answer Type Questions)

Note : Attempt any two questions. Each question carries 12 marks. $2 \times 12 = 24$

1. Define hybridization of orbitals. Name different kinds of hybridization and the shapes of molecules obtained from such hybridizations.

2. Starting from Arrhenius equation derive the expression :

$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303} \left[\frac{1}{T_1} - \frac{1}{T_2} \right]$$

3. Discuss the structure and shape of carbocation and carbanion.
4. Give the preparation and important uses of the Teflon.

Section—C

(Long Answer Type Questions)

Note : Attempt any *one* question. Each question carries 20 marks. $1 \times 20 = 20$

1. (a) Explain the following terms with suitable examples :

- (i) Plane of symmetry
- (ii) Axis of symmetry
- (iii) Centre of symmetry

- (b) Draw the planes with Miller indices (112), (210).

2. (a) Discuss the *two* methods use for the determination of order of reaction.

(b) Derive Nernst equation for measuring EMF of a cell.

3. (a) Give the mechanisms of the cannizaaro reaction.

(b) Write a short note on optical isomerism.

4. (a) Describe Redox titrations with examples.

(b) Discuss water treatment methods for boiler feed water.

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CY-1844

**B. Tech. (First Semester)
EXAMINATION, 2024-25
ENGINEERING CHEMISTRY**

Time : Two Hours

Maximum Marks : 100

Note : Attempt questions from all Sections as directed.

Section—A

(Very Short Answer Type Questions)

Note : Attempt any *seven* questions. Answer each question in maximum 30 words. Each question carries 8 marks. $7 \times 8 = 56$

1. State *three* main points of valence bond theory.
2. Give *three* characteristics of molecular crystals.

P. T. O.

3. Define the order of the reaction.
4. Explain the term 'Phase'.
5. Define electrofile and nucleophile.
6. Name *two* factors that determine the stability of Carbocation.
7. What are synthetic fibers ? Give *two* examples.
8. What are biodegradable polymers ?
9. Explain the term Auxochromophores.
10. What is Biomass ?

Section—B

(Short Answer Type Questions)

Note : Attempt any *two* questions. Answer each question in maximum **150** words. Each question carries 12 marks.

$$2 \times 12 = 24$$

1. With the help of a diagram, indicate various forces of attraction and repulsion in a molecule of H_2 .

2. Define activation energy. How is it determined graphically using Arrhenius equation ?
3. What is a free radical ? Discuss its stability.
4. Give the preparation and important uses of Nylon-66.

Section—C

(Long Answer Type Questions)

Note : Attempt any *one* questions. Answer each question in maximum 300 words. Each question carries 20 marks.

$$1 \times 20 = 20$$

1. (a) Represent the following diagrammatically :
 - (i) Body-centred cubic lattice
 - (ii) Face-centred orthorhombic lattice
 - (iii) Hexagonal lattice
 - (iv) Simple tetragonal lattice
- (b) What is meant by total elements of symmetry ?

2. (a) What is a second order reaction ?
Derive an equation given the velocity constant of a second order reaction.
- (b) What do you understand by the EMF of the cell- ? How can it be measured experimentally ?
3. (a) Give the mechanism of the Diels-Alder reaction.
- (b) Explain E-Z nomenclature.
4. (a) Discuss acid-base titrations.
- (b) Give the classification of fuels.

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B. Tech. (First Semester)

EXAMINATION, 2023-24

ENGINEERING CHEMISTRY

Time : Two Hours

Maximum Marks : 100

Note : Attempt questions from both Sections as directed.

Section—A

(Short Answer Type Questions)

Note : Attempt any *five* questions. Each question carries 8 marks. $5 \times 8 = 40$

1. Compare the bond order of O_2 , O_2^+ and O_2^{2+} .
2. Define the band theory.
3. What is Crystal lattice and Lattice energy ?

P. T. O.

4. Calculate the density of diamond from the fact that it has FCC structure with atoms per lattice point and a unit cell of edge $3.569 \text{ \AA}$.
5. A zero order reaction is 50% completed in 20 minutes. What percentage would be completed at the end of 30 minutes ?
6. Give a brief note on activation energy and catalysis.
7. Write important features of a galvanic cell.
8. Define the inductive effect with a suitable example.
9. Write differences between carbonium and carbanion ion.
10. Discuss the E/Z nomenclature.
11. What are biopolymers ?
12. Write any two applications of organo-aluminium compounds.
13. What do you know about IR spectroscopy ?

14. How is exhausted ion exchange resin regenerated ?
15. Distinguish between gross and net calorific value of a fuel.

Section—B

(Long Answer Type Questions)

Note : Attempt any *two* questions. Each question carries 30 marks.

$$2 \times 30 = 60$$

1. (a) Draw the molecular orbital diagram of CO.
- (b) What is the X-ray crystallography ?
2. (a) Derive an equation for rate constant of first order reaction.
- (b) Discuss the structure of graphite and comment on its conducting nature.
3. (a) Derive the Nernst equation.
- (b) Give the mechanism of aldol condensation.

4. (a) Write the applications of following polymers :

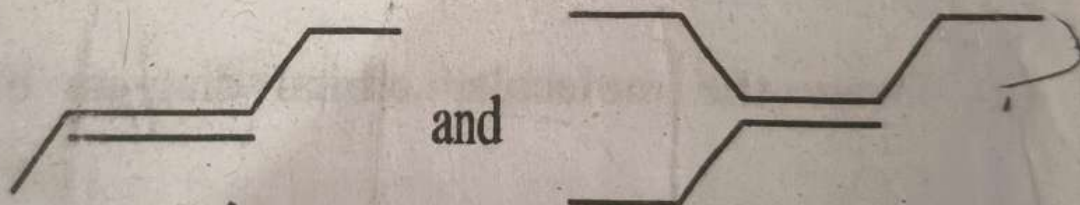
(i) Nylon 6

(ii) Conducting polymers

(iii) Phenol formaldehyde resins

(iv) PMMA.

(b) Write E and Z configuration of the following compounds :



5. (a) Write the applications of ultraviolet spectroscopy.

(b) Write any *three* applications of Grignard's reagent.